A Bialgebra Framework for Positivity in Schubert Calculus

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CombinaTexas 2026

2026

Schubert polynomials

- Multivariate polynomials representing cohomology classes of Schubert varieties in the complete flag variety.
- Indexed by permutations, with the Schubert polynomial $\mathfrak{S}_w$ corresponding to a permutation w.
- Generalize the Schur polynomials, so their structure constants generalize the Littlewood-Richardson coefficients.
- Structure constants are nonnegative integers, but no positive combinatorial formula is known for them in general.

$$\mathfrak{S}_{41523}(x_1, x_2, x_3) = x_1^3 x_2^2 + x_1^3 x_2 x_3 + x_1^3 x_3^2$$

What is a positive formula?

- Trying to find such a formula has made generations of mathematicians question their sanity
- What do we mean when we say a "positive formula?"
- Canditate: complexity class $\#\mathrm{P}$. The number of "yes" answers to an NP problem.

Theorem (S, 2026+)

Schubert polynomial multiplication is in $\#\mathrm{P}$.

Why is this not a positive formula?

- Rearranges the problem so that the cancellation results in a 0-1 valued function, without combinatorially resolving the cancellation.

A run of the mill bialgebra

Define

$$\mathbf{S} = \bigoplus_{n \geq 0} \mathbf{S}_n, \qquad \mathbf{S}_n = \mathbb{Q}[x_1, \ldots, x_n].$$

Multiplication in $\mathbf{S}$ is componentwise:

- if $f, g \in \mathbf{S}_n$, then $f \cdot g$ is the usual product in $\mathbb{Q}[x_1, \ldots, x_n]$;
- if $f \in \mathbf{S}_m$, $g \in \mathbf{S}_n$ with $m \neq n$ and $m, n > 0$, then $f \cdot g = 0$.

So each graded piece is a familiar polynomial ring, and cross-component products are trivial. The coproduct is

$$\Delta(x_c) = \sum_{ab=c} x_a \otimes x_b$$

where ab is the concatenation of weak compositions. This makes $\mathbf{S}$ into a graded bialgebra.

$$\Delta(x_1^2 x_3 x_5) = 1 \otimes x_1^2 x_3 x_5 + x_1^2 \otimes x_3 x_5 + x_1^2 x_3 \otimes x_5 + x_1^2 x_3 x_5 \otimes 1.$$

Double grading

In addition to the degree as a polynomial, $\mathbf{S}$ is also graded by inclusion in the direct summand $\mathbf{S}_n$, which we as interpret as length.
We define the dual algebra $\mathcal{DS}$ to be the graded dual of $\mathbf{S}$ with respect to this $\mathbb{N} \times \mathbb{N}$ grading. For the monomial basis, which is indexed by weak compositions, we define the dual basis $[\alpha]$ for weak compositions α by

$$\langle x_\alpha, [\beta] \rangle = \delta_{\alpha,\beta}$$

Hence $\mathcal{DS}$ is also graded by length and degree, with the latter being the sum of the composition, and the product and coproduct are dual to those of $\mathbf{S}$ with respect to the degree grading.

Dual indexing and product

We have that the dual product is concatenation:

$$[\alpha] \cdot [\beta] = [\alpha\, \beta]\,.$$

Hence $\mathcal{DS}$ is a free associative algebra (tensor algebra on the generators $[k]$, $k \geq 0$). The coproduct is

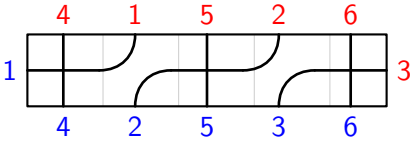
$$\Delta([\gamma]) = \sum_{\alpha+\beta=\gamma} [\alpha] \otimes [\beta]\,.$$

We now define and investigate a curious combinatorially defined basis of $\mathcal{DS}$ indexed by code-length bounded permutations.

Crossing diagrams

Think of a one-row strip of $n-1$ boxes.

- Label the left edge by 1.
- Label the bottom by a permutation of the numbers $2, 3, \ldots, n$ (left to right).
- In each box, place either a *cross* ⊞ or a *bump* ◪.



Follow the strands through the row. The admissibility condition is that it cannot undo inversions, only create new ones.

A nice combinatorial basis

Let Ξ_w^k be the basis indexed by permutations w such that $w(p) < w(p+1)$ for all $p > k$ satisfying the following:

- $\Xi_w^1 = [w(1) - 1]$
- For a nonnegative integer $[p]$, we have that

$$[p] \cdot \Xi_w^k = \sum_{w'} \Xi_{w'}^{k+1}$$

  where w' is obtained from w through a valid crossing diagram with p crosses after adding 1 to each element of w and adding a new first element 1.

I can provide an explicit (but mildly cancellative) formula for Ξ_w^k in the word basis, as well as a positive formula for the product of Ξ_u^p and Ξ_v^q in terms of crossing diagrams. The coproduct structure constants are also nonnegative integers, but I do not have a positive combinatorial formula for them yet...

Taking off the mask: The dual Schubert basis

The basis satisfies

$$\langle \mathfrak{S}_u(x_1, \ldots, x_n), \Xi_v^n \rangle = \delta_{u,v}$$

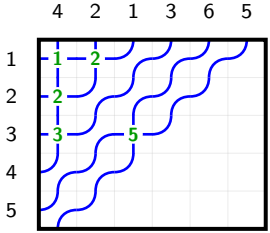
where $\mathfrak{S}_u(x_1, \ldots, x_n)$ is the Schubert polynomial corresponding to the permutation u. The coefficients $c_{u,v}^w$ of the coproduct formula

$$\Delta\, \Xi_w^n = \sum_{u,v} c_{u,v}^w\, \Xi_u^n \otimes \Xi_v^n$$

are the Littlewood-Richardson coefficients for Schubert polynomials!

Bounded pipe dreams

Pipe dreams are well-studied objects introduced in the 1990s by Bergeron and Billey to give a nicer combinatorial formula for Schubert polynomials equivalent to an earlier one by Billey, Jockusch, and Stanley involving reduced words/compatible sequences. Bounded pipe dreams have an additional length parameter tacked on.



Theorem (Bergeron, Billey, 1993)

The Schubert polynomial $\mathfrak{S}_w$ is equal to

$$\sum_{R \in \mathrm{Pipe}(w)} x_{\mathrm{wt}(R)}$$

Transition algorithm on bounded pipe dreams

There is an algorithm that takes a bounded pipe dream of length n with row n empty to a unique pipe dream of length $n-1$ in a way that matches the Schubert transition formula in a strong, weight-preserving, bijective form.

Input: A bounded pipe dream R with $\mathrm{length}(R) = n$ and row n empty.
Output: A bounded pipe dream R' with $\mathrm{length}(R') = n-1$, $|R'| = |R|$ (same number of crossing in each row), and $w_R \overset{n}{\searrow} w_{R'}$.
if R with last row removed is a valid bounded pipe dream **then**
    return R with last row removed
else
    Locate crossings corresponding to maximal right descents of $w(R)$ until the maximal right descent is $n-1$, say at rows $i_1 < i_2 < \cdots < i_p$ with corresponding columns $j_1, j_2, \ldots, j_p$.
    $R^- \leftarrow R \setminus \{(i_1, j_1), \ldots, (i_p, j_p)\}$.
    Let $I \leftarrow (i_1, i_2, \ldots, i_p)$.
    $D \leftarrow R^- \cup \{(n,1), (n,2), \ldots, (n,p)\}$.
    $D' \leftarrow I \overset{n-1}{\rightsquigarrow} D$ (an insertion algorithm similar to existing ones for pipe dreams)
    $R' \leftarrow \{(i,j) \in D' \mid i < n\}$ as a bounded pipe dream with $n-1$ rows.
    return R'.
end if

Bounded pipe dream ring product

Given bounded pipe dreams R_1, R_2, define $R_1 \star R_2$ by taking the sum of all R obtained by

- *untransitioning* R_1 in all possible valid ways to create space while remaining a pipe dream,
- shifting R_2 upward,
- placing the shifted R_2 into the new empty region, unchanged, if doing so yields a valid pipe dream.

This allows us to lift the product of the dual algebra $\mathcal{DS}$ to a product on formal linear combinations of bounded pipe dreams.

Theorem (S, 2026+)

The product $\star$ is associative and gives a well-defined ring structure on the vector space Pipe *spanned by bounded pipe dreams.*

Pipe dream ring as a lift of $\mathcal{DS}$

There are two surjective linear maps from the pipe dream ring $\mathcal{R}$ to the dual algebra $\mathcal{DS}$:

$$\pi_{\mathrm{Sch}} : \mathcal{R} \twoheadrightarrow \mathcal{DS}, \qquad \pi_{\mathrm{Sch}}(G) = \Xi^{(n)}_{\mathrm{perm}(G)},$$

$$\pi_{\mathrm{wt}} : \mathcal{R} \twoheadrightarrow \mathcal{DS}, \qquad \pi_{\mathrm{wt}}(G) = [\mathrm{wt}(G)]\,.$$

By virtue of the fact that $\mathcal{DS}$ is free, there is a section $\iota : \mathcal{DS} \hookrightarrow \mathcal{R}$ of π_{wt} that sends $[i]$ for an integer i to the unique pipe dream r_i with $\mathrm{wt}(r_i) = (i)$. It turns out that these are all ring homomorphisms and ι is also a section of π_{Sch}.

Theorem (S, 2026+)

We have that π_{Sch} and π_{wt} are surjective ring homomorphisms to $\mathcal{DS}$ and ι is an injective ring homomorphism from $\mathcal{DS}$ to $\mathcal{R}$ such that

$$\pi_{\mathrm{Sch}} \circ \iota = \pi_{\mathrm{wt}} \circ \iota = \mathrm{id}_{\mathcal{DS}}.$$

Littlewood-Richardson rule for (dual) Schubert polynomials

We immediately obtain a formula for the product of dual Schubert elements.

Theorem (S, 2026+)

For dual Schubert elements $\Xi_u^{n_1}$, $\Xi_v^{n_2}$, we have

$$\Xi_u^{n_1} \cdot \Xi_v^{n_2} = \sum_w d_{u,v}^w(n_1, n_2)\, \Xi_w^{n_1+n_2},$$

where, fixing pipe dreams R_u for u with $\mathrm{length}(R_u) = n_1$ and R_v for v with $\mathrm{length}(R_v) = n_2$, $d_{u,v}^w(n_1, n_2)$ is the number of bounded pipe dreams $R_w \in \mathrm{Pipe}(w)$ such that R_w has a coefficient of 1 in the product $R_u \star R_v$.
Equivalently,

$$\mathfrak{S}_w(x_1, \ldots, x_{n_1+n_2}) = \sum_{u,v} d_{u,v}^w(n_1, n_2)\, \mathfrak{S}_u(x_1, \ldots, x_{n_1})\mathfrak{S}_v(x_{n_1+1}, \ldots, x_{n_1+n_2}).$$

Interpretation of π_{Sch} as a basis transition

π_{Sch} can be interpreted as a transition from the word basis of $\mathcal{DS}$ to the Schubert basis. In particular, we have that

Theorem (S, 2026+)

Let $D \in \mathcal{DS}$ be an element such that

$$D = \sum_{\alpha} c_\alpha [\alpha]$$

where α ranges over all weak compostiions. Then

$$D = \sum_{R \in \mathrm{Pipe}, \mathtt{wt}(R) = \alpha} c_\alpha \pi_{\mathrm{Sch}}(R)$$

Homomorphism of rings into the direct product

π_{Sch} can be interpreted as a transition from the word basis of $\mathcal{DS}$ to the Schubert basis. In particular, we have that

Theorem (S, 2026+)

There is a homomorphism $\Phi : \mathcal{DS} \times \mathcal{DS}$ given by

$$\Phi(D) = (\pi_{\mathrm{Sch}}(D), \pi_{\mathrm{wt}}(D))$$

such that if D is in the subalgebra of Pipe generated by the single rows, then $\Phi(D) = (d, d)$ for some $d \in \mathcal{DS}$.

Positivity of Schubert polynomial multiplication (!!!)

General positivity theorem

Examples

This applies to the following examples of equivalence relations on reduced words:

- Schubert polynomials: $\mathrm{perm}(W) = \mathrm{perm}(W')$.
- Fundamental slide polynomials ($W \equiv W'$ iff $W = W'$)
- Key polynomials: $W \equiv W'$ if W and W' are Coxeter-Knuth equivalent.

Forest polynomials

Forest polynomials $\mathfrak{P}_F$ are a more recently discovered basis of $\mathbf{S}$ in which the Schubert basis also expands positively. They are indexed by objects F known as "indexed forests," which are canonically labeled forests of binary trees. Given a word W, there is an insertion algorithm that yields a labeled indexed forest denoted $\Omega(W)$. We then say that $\Omega(R)$ has shape F if the indexed forest obtained from R by this insertion algorithm is F.

Theorem (S, 2026+)

Let F be an indexed forest with $\mathfrak{c}(F) = \mathfrak{c}(w)$ for a permutation w. Then

$$\mathfrak{P}_F = \sum_{\substack{R \in \mathrm{Pipe}(w) \\ \Omega(R) \text{ has shape } F}} \mathrm{wt}(R)$$

Dual product of forest polynomials

Theorem (S, 2026+)

For each positive integer k, forest polynomials with $n \geq k$ variables satisfy the formula

$$\mathfrak{P}_F = \sum_{\gamma} d^F_{G,H}(k)\, \mathfrak{P}_G(x_1, \ldots, x_k) \mathfrak{P}_H(x_{k+1}, \ldots, x_n),$$

Where, fixing bounded pipe dreams R_G, R_H with $\mathrm{length}(R_G) = k$, $\mathrm{length}(R_H) = n - k$, $\mathrm{shape}(\Omega(R_G)) = G$ and $\mathrm{shape}(\Omega(R_H)) = H$, $d^H_{F,G}(k)$ is the number of bounded pipe dreams R with $\mathrm{length}(R) = n$ and $\mathrm{shape}(\Omega(R)) = F$ such that R has a coefficient of 1 in the product $R_G \star R_H$.